



Comparative Study of Data Distribution Before and After Outlier Treatment

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Introduction

Problem

- Outliers can distort mean and variance.
- Extreme values can create skewness.
- Visual patterns may be misleading.

Approach

- Inspect the original distribution.
- Detect outliers using the IQR method.
- Treat, recalculate, and compare.

Why now?

- Cleaner data improves statistical interpretation.
- Distribution becomes easier to interpret.
- Supports reliable downstream modeling.

Dataset

Source Kaggle / Sample Superstore (Sales Data)
Records 9,994
Features Sales, Profit, Quantity, Discount
Other Valid observations retained for analysis

Data preparation

- Handle missing values and outliers
- Apply IQR-based outlier treatment
- Compare distributions before and after

Objective

Assess the effect of treatment

OUTLIER TREATMENT PIPELINE

1

Inspect

2

IQR Detect

3

Treat

4

Recalculate

5

Compare

$IQR = Q3 - Q1$ | Lower bound = $Q1 - 1.5 \times IQR$ | Upper bound = $Q3 + 1.5 \times IQR$

TOOLS & ASSESSMENT

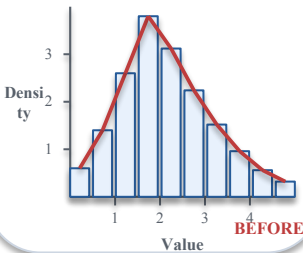
- Python
- pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy
- Spread
- Skewness
- Mean
- Median
- IQR

KEY FINDINGS

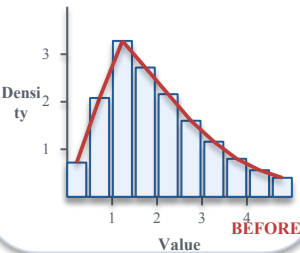
- Extreme observations have stronger influence before treatment.
- Median and IQR remains robust references.
- Explore Isolation Forest, DBSCAN, model impact and other
- Real-world datasets

RESULTS & COMPARATIVE ANALYSIS

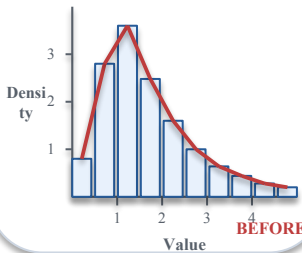
Sales



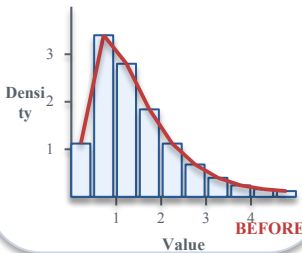
Profit



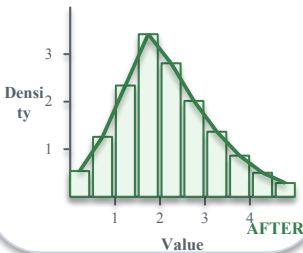
Quantity



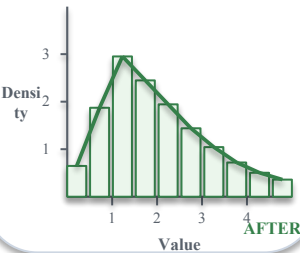
Discount



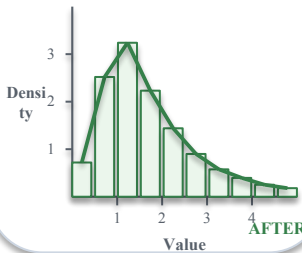
Sales



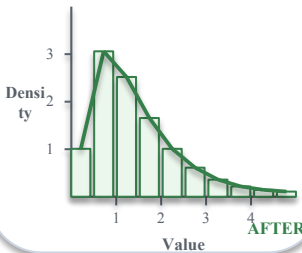
Profit



Quantity



Discount



VISUAL READ

Before: greater spread and tail influence

After: more stable central pattern

Clearer interpretation

DISTRIBUTION COMPARISON SUMMARY

Variable	Mean Before	Mean After	SD Before	SD After
Sales	229.86	223.12	623.25	339.14
Profit	28.66	20.48	234.26	91.36
Quantity	3.79	3.68	2.22	1.77
Discount	0.16	0.15	0.21	0.14

Variable	Before	After
Sales	4.21	0.92
Profit	5.63	0.74
Quantity	0.76	0.21
Discount	1.19	0.18

Reference values show lower spread and skewness after treatment across all four variables.

CONCLUSION

- Outlier treatment can reduce distortion and reveal a more representative distribution when applied reproducibly.